

TITAN molecular prediction report

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Research-use molecular profile

COAD - COAD example B

Models applied	Continuous	Binary	Slides pooled
20	10	10	1

This report applies models trained in TCGA and **not externally validated**. It is a discovery/research output, not a diagnostic report. Binary LDA scores are uncalibrated and must not be read as probabilities.

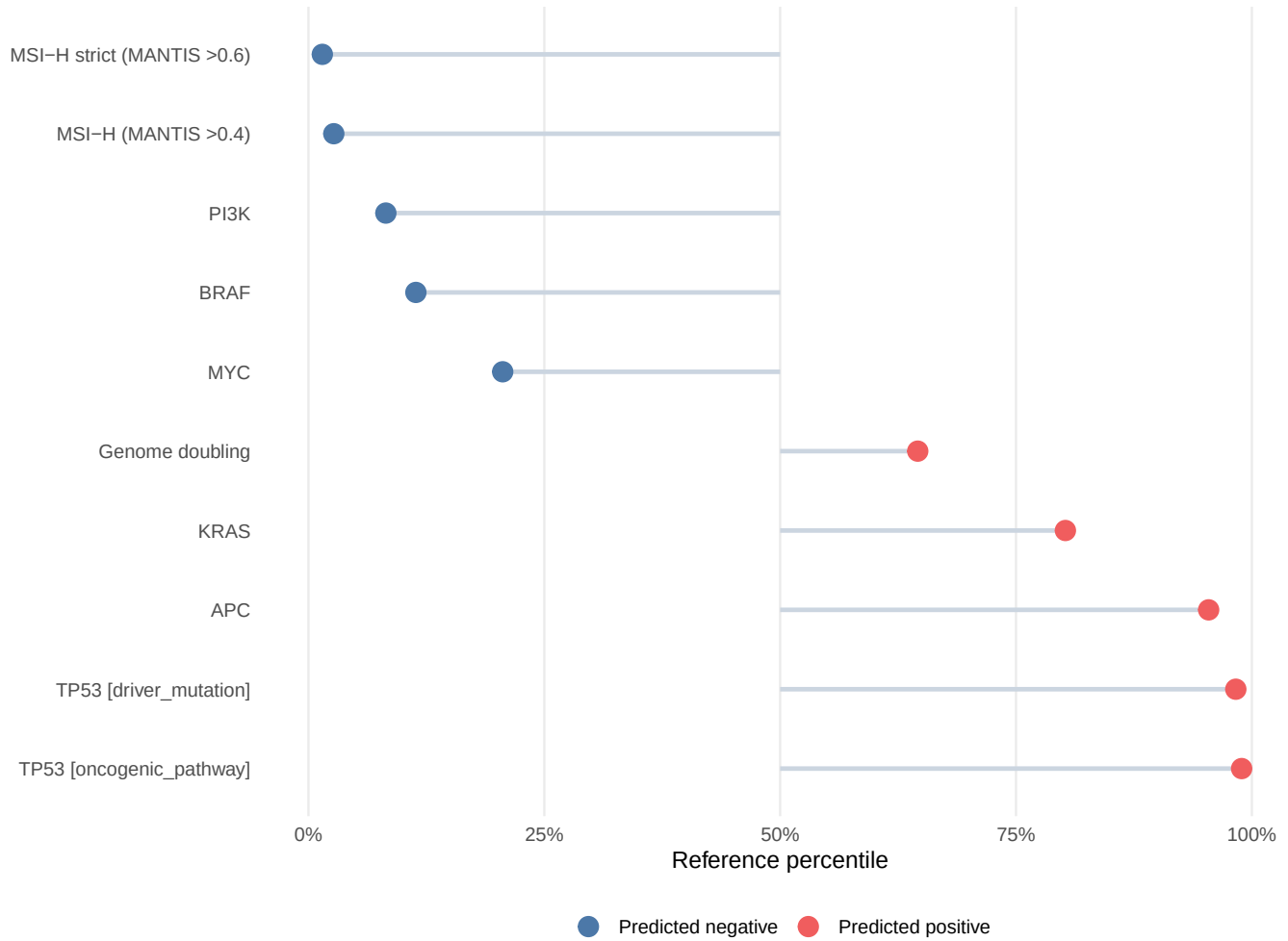
Continuous molecular profile

Percentile relative to repeated TCGA out-of-fold predictions



Binary molecular calls

LDA-score percentile; this is not a calibrated probability



Continuous predictions

	Family	Endpoint	Prediction	Reference percentile	Evidence tier
16	thorsson	Lymphocyte Infiltration Signature Score	-1.34	0.1%	B
20	thorsson	TIL Regional Fraction	-0.7451	0.1%	A
14	microsatellite_instability	MSIsensor score	-5.113	0.1%	B
13	microsatellite_instability	MANTIS score	0.1752	0.1%	B
18	thorsson	SNV Neoantigens	3.377	0.1%	B
17	thorsson	Nonsilent Mutation Rate	0.9799	0.1%	B
19	thorsson	Silent Mutation Rate	0.4191	0.1%	B
12	aneuploidy	Deleted arm count	12.21	99.6%	B
15	thorsson	Aneuploidy Score	16.96	99.6%	B
11	aneuploidy	Aneuploidy score	18.25	98.2%	B

Binary predictions

Endpoint	Predicted class	LDA score	Score percentile	Evidence tier
Genome doubling	1	0.2739	64.6%	B
APC	1	3.232	95.4%	A
BRAF	0	-9.303	11.4%	B
KRAS	1	0.7263	80.2%	B
TP53 [driver_mutation]	1	3.526	98.3%	A
MSI-H (MANTIS >0.4)	0	-9.711	2.7%	A
MSI-H strict (MANTIS >0.6)	0	-16.94	1.5%	A
MYC	0	-2.89	20.6%	B
PI3K	0	-2.294	8.2%	B
TP53 [oncogenic_pathway]	1	4.146	98.9%	B

Quality and provenance

- Patient-level aggregation: arithmetic mean across all supplied slide vectors.
- Maximum fraction outside endpoint-specific TCGA training ranges: 0.1%.
- TITAN input schema: 768 named features (`titan_000`–`titan_767`).
- Decomposition: seeded rSVD in fastPLS; binary classification: PLS-LDA.
- Intended use: research only; TCGA discovery atlas without independent external validation.